

Practice Exam 3
Sixty-five multiple-choice and multiple-response
questions, timed to the length of the real examination.

QUESTIONS	TIME LIMIT	EXAM
65 6 multiple-response	90 minutes · 83 s per question	3 / 3 no question is repeated

B E F O R E Y O U S T A R T

- Set a timer for 90 minutes and work without notes,
Q %
documentation, or a browser.
12 18
- Record every answer on the response sheet near the
e AI 17 26
back. Do not write in the question pages if you plan to
retake this exam.
19 29
- Questions marked S E L E C T T W O require exactly two
answers. There is no partial credit – both must be right.
e AI 8 12
- Nothing is deducted for a wrong answer, so answer every
question. Flag the ones you guessed and revisit them if
9 14
time allows.
65 100
- When the timer ends, stop, then score yourself with the
answer key and read the explanation for every question
you missed.

THIS EXAM COVERS

- DOMAIN
- D1 Fundamentals of AI and ML
- D2 Fundamentals of Generativ
- D3Applications of Foundation
- Models
- D4 Guidelines for Responsibl
- D5 Security, Compliance, and
Governance

Total

P A C I N G C H E C K P O I N T S

Q17 after 23 min	Q33 after 45 min	Q49 after 68 min	Q65 at 90 min
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10 questions cover the revised objectives: MCP, AgentCore, context engineering, distillation, Kiro, Strands, LLM-as-a-judge.
Exam 3 of 3

- 1 A developer wants Kiro to access an external issue tracker and internal API through standardized tool interfaces. Which Kiro capability supports this requirement?
- A MCP server configuration
 - B Diffusion sampling
 - C SageMaker Model Monitor
 - D ROUGE evaluation
- 2 A company wants to implement fully managed support for end-to-end Retrieval Augmented Generation (RAG) workflow in Amazon Bedrock. What do you recommend?
- A Continued pretraining in Amazon Bedrock
 - B Watermark detection for Amazon Bedrock
 - C Guardrails for Amazon Bedrock
 - D Knowledge Bases for Amazon Bedrock
- 3 Which of the following is a generative AI-powered assistant that can answer questions, provide summaries, generate content, and securely complete tasks based on data and information in the enterprise systems?
- A Amazon Q in Connect
 - B Amazon Q in QuickSight
 - C Amazon Q Business
 - D Amazon Q Developer
- 4 Which of the following represents the correct statement regarding Amazon Q Developer?
- A Amazon Q Developer can neither be used in the integrated development environments (IDEs) nor the AWS Management Console
 - B Amazon Q Developer can be used in integrated development environments (IDEs) as well as the AWS Management Console
 - C Amazon Q Developer can only be used in the AWS Management Console
 - D Amazon Q Developer can only be used in the integrated development environments (IDEs)

- 5 In foundation-model distillation, what are the roles of the teacher and student models?
- A The teacher is the smaller model being optimized, and the student labels the training data manually
 - B The teacher is a larger, more capable model whose knowledge is transferred to a smaller, faster student model
 - C Both models must have identical size, latency, and cost
 - D The student creates embeddings, and the teacher stores them in a vector database
- 6 What is the key difference between multi-class classification and multi-label classification in machine learning?
- A Multi-class classification does not require labeled data, whereas multi-label classification requires labeled data for training
 - B Multi-class classification is used exclusively for image data, whereas multi-label classification is used exclusively for text data
 - C Multi-class classification allows each instance to belong to multiple classes simultaneously, whereas multi-label classification restricts each instance to one class only
 - D Multi-class classification assigns each instance to one of several possible classes, while multi-label classification assigns each instance to one or more classes
- 7 Which of the following highlights the differences between model parameters and hyperparameters in the context of generative AI?
- A Both Hyperparameters and model parameters are values that define a model and its behavior in interpreting input and generating responses
 - B Hyperparameters are values that define a model and its behavior in interpreting input and generating responses. Model parameters are values that can be adjusted for model customization to control the training process
 - C Model parameters are values that define a model and its behavior in interpreting input and generating responses. Hyperparameters are values that can be adjusted for model customization to control the training process
 - D Both Hyperparameters and model parameters are values that can be adjusted for model customization to control the training process

- 8 A company is using Amazon Bedrock and it wants to set an upper limit on the number of tokens returned in the model's response. Which of the following inference parameters would you recommend for the given use case?
- A Response length
 - B Top P
 - C Top K
 - D Stop sequence
- 9 Which feature of Amazon Bedrock allows developers to automate repetitive tasks in their AI applications?
- A Predefined templates
 - B Serverless environment
 - C Model access permissions
 - D Use of agents
- 10 An ecommerce company is transitioning to AWS Cloud and wants to use Amazon Bedrock for product recommendations. The company wants to provide its own labeled training dataset to improve the selected Foundation Model's (FM) performance. Which of the following represents the best-fit solution for the given use case?
- A Leverage Amazon Bedrock to make a separate copy of the base FM model and train this private copy of the model using the labeled training dataset
 - B Leverage Amazon Bedrock to make a public copy of the base FM model and train this public copy of the model using the labeled training dataset
 - C Leverage Amazon Bedrock to discard the selected FM and create a new model from scratch by using the labeled training dataset
 - D Leverage Amazon Bedrock to train the base FM itself using the labeled training dataset
- 11 Which of the following can assist IT professionals in coding, testing, and upgrading applications?
- A Amazon Q in Connect
 - B Amazon Q Developer
 - C Amazon Q Business
 - D Amazon Q in QuickSight

- 12 What is a key difference between structured data and unstructured data in the context of Machine Learning?
- A Structured data is typically freeform text that lacks any specific format, whereas unstructured data is organized in a tabular format with rows and columns
 - B Structured data is used exclusively for training machine learning models, whereas unstructured data is used solely for storing information without any analytical purpose
 - C Structured data includes data like text, images, and videos, whereas unstructured data is limited to numerical data only
 - D Structured data is organized in a predefined manner, often in rows and columns, making it easy to search and analyze, while unstructured data lacks a specific format and includes data like text, images, and videos
- 13 Which pricing model is suitable for large and steady workloads in Amazon Bedrock?
- A Provisioned throughput mode
 - B Subscription-based pricing with monthly contracts
 - C On-demand pricing based on input and output tokens
 - D Pay-as-you-go with fixed monthly fee
- 14 What is one of the primary challenges in machine learning implementation?
- A Limited applications of machine learning in real-world scenarios
 - B Insufficient computational power to run basic machine learning models
 - C Difficulty in collecting and preparing high-quality data for training models
 - D Lack of available machine learning algorithms
- 15 What is the primary function of a prompt in the context of large language models?
- A To serve as the input or query that generates a response from the model
 - B To reduce the size of the model for better efficiency
 - C To train the model with new data
 - D To provide the output desired from the model

- 16 Which feature of Amazon Q Business allows the AI assistant to interact with third-party applications like Jira and Salesforce?
- A Plugins
 - B Guardrails and controls
 - C Prebuilt connectors
 - D Retrieval augmented generation (RAG)
- 17 A company is creating a custom search solution that will bring together the company's data repositories, FAQs, and support tickets. The support tickets might contain personally identifiable information (PII) that needs to be redacted before the tickets are processed to create the search indexes. Which AWS service will help you redact the PII in support tickets?
- A Amazon Lex
 - B Amazon Comprehend
 - C Amazon Kendra
 - D Amazon Textract
- 18 What is the primary benefit of using Amazon Bedrock for AI and machine learning tasks?
- A It automates the process of data labeling and augmentation for machine learning models.
 - B It enables real-time speech recognition and transcription capabilities.
 - C It provides a fully managed infrastructure for training custom AI models from scratch.
 - D It allows users to access and use pre-trained foundation models from various AI model providers without needing to manage infrastructure.
- 19 What is the primary purpose of AWS AI service cards?
- A To serve as a marketplace for purchasing third-party AI services using pre-paid cards
 - B To provide detailed technical documentation for setting up AWS AI services
 - C To offer transparency and information about the intended use, limitations, and potential impacts of AWS AI services, helping users implement Responsible AI practices
 - D To provide a platform for users to share their AI models and datasets with the community

- 20 What is a key role of agents in generative AI applications?
- A To act as intermediaries between the AI model and business backend systems
 - B To replace the AI model in understanding language
 - C To perform manual data entry tasks
 - D To only generate responses without interacting with other systems
- 21 Which of the following scenarios best illustrates human bias in the context of machine learning?
- A A machine learning model trained on historical hiring data consistently recommends male candidates for technical roles
 - B An automated translation service frequently makes errors when translating idiomatic expressions between languages
 - C A machine learning algorithm predicts customer churn based on historical data, but the data is skewed due to seasonal trends
 - D A data scientist selects features for a machine learning model based on their personal beliefs about which attributes are important, leading to a biased model
- 22 What is a key difference between Foundation Models (FMs) and Large Language Models (LLMs) in the context of generative AI?
- A Foundation Models are specifically designed for text generation, while Large Language Models can generate images, videos, and audio
 - B Foundation Models are only used in academic research, while Large Language Models are used in commercial applications
 - C Large Language Models are pre-trained on massive datasets and can be fine-tuned for specific tasks, whereas Foundation Models are not pre-trained and are built from scratch for each application
 - D Foundation Models serve as a broad base for various AI applications by providing generalized capabilities, whereas Large Language Models are specialized for understanding and generating human language
- 23 A bank wants refund limits to be enforced even if an agent is manipulated by a malicious prompt. Which design provides the strongest deterministic control?
- A Describe the refund limit only in the agent's system prompt
 - B Increase the model's temperature before calling the refund tool
 - C Attach an AgentCore policy engine with a Cedar policy to the Gateway
 - D Store the refund policy in the agent's short-term memory

- 24 How does the inference parameter Top K influence the model response for Amazon Bedrock?
- A Influences the percentage of most-likely candidates that the model considers for the next token
 - B Specifies the sequences of characters that stop the model from generating further tokens
 - C Influences the number of most-likely candidates that the model considers for the next token
 - D Influences the likelihood of the model selecting lower-probability outputs, thereby impacting the creativity of the model's output
- 25 Which statement best describes Amazon Bedrock AgentCore?
- A It is a single proprietary agent framework that requires a specific foundation model
 - B It is a modular platform for securely building, deploying, and operating agents at scale with different frameworks and models
 - C It is a vector database available only to Amazon Bedrock Knowledge Bases
 - D It is a service used only for manually labeling model-training data
- 26 What is the primary purpose of vectors in large language models?
- A To generate high-quality images
 - B To split text into tokens
 - C To represent the numerical meaning of words
 - D To compress input data into a latent space
- 27 In Amazon Bedrock AgentCore Gateway, what does a gateway target represent?
- A The foundation model that must make every tool-selection decision
 - B The backend tool or resource and the configuration Gateway uses to connect to it
 - C A dataset containing expected answers for model evaluation
 - D A long-term memory strategy for retaining user preferences

- 28 The marketing department at a retail company is working on a campaign for the upcoming Thanksgiving holidays. The department wants to exclude competitive brand names or sensitive topics from the content produced via generative AI prompts. What type of prompting technique does the given use case represent?
- A Few-shot Prompting
 - B Chain-of-thought prompting
 - C Negative prompting
 - D Zero-shot Prompting
- 29 What is the key difference between machine learning and artificial intelligence?
- A Machine learning encompasses the broader concept of artificial intelligence, which includes rule-based systems and decision-making processes
 - B Machine learning is a subset of artificial intelligence that involves training algorithms to learn from data, while artificial intelligence encompasses a wider range of technologies aimed at simulating human intelligence
 - C Artificial intelligence is a subset of machine learning that focuses solely on statistical analysis
 - D Artificial intelligence is concerned only with physical robots, while machine learning focuses exclusively on software algorithms
- 30 How can you prevent model-overfitting in machine learning?
- A By increasing the complexity of the model to ensure it captures all nuances in the training data
 - B By avoiding any form of model validation or testing to prevent the model from learning incorrect patterns
 - C By only training the model on a small subset of the available data to reduce the amount of information it has to learn
 - D By using techniques such as cross-validation, regularization, and pruning to simplify the model and improve its generalization
- 31 Which task ensures the confidentiality and integrity of data at rest and in transit in AI systems?
- A Infrastructure protection
 - B Vulnerability management
 - C Threat detection
 - D Data encryption

- 32 Which AWS service offers flexibility to include human input across the Machine Learning lifecycle to improve the accuracy and relevancy of the models?
- A Amazon SageMaker Clarify
 - B Amazon SageMaker Role Manager
 - C Amazon SageMaker Ground Truth
 - D Amazon SageMaker Feature Store
- 33 A company wants its agents to use existing AWS Lambda functions and REST APIs without rewriting each integration for a different agent framework. Which solution best meets this requirement?
- A Create an Amazon Bedrock model-distillation job
 - B Add the functions and APIs as targets behind an Amazon Bedrock AgentCore Gateway
 - C Store the API responses in SageMaker Model Cards
 - D Increase the model temperature so that it can infer the API schema
- 34 A business needs an automated solution that can extract text from thousands of receipts and invoices generated across all its stores. Which AWS Machine Learning (ML) service can offer the most optimal solution for this use case ?
- A Amazon Comprehend
 - B Amazon Transcribe
 - C Amazon Rekognition
 - D Amazon Textract
- 35 Which Amazon SageMaker service will help understand how an input feature contributes to the predictions of a machine learning model?
- A Amazon SageMaker Ground Truth
 - B Amazon SageMaker Canvas
 - C Amazon SageMaker JumpStart
 - D Amazon SageMaker Clarify

- 36 Which capabilities are supported by the Strands Agents SDK? S E L E C T T W O
- A Connecting to tools through MCP
 - B Building multi-agent patterns such as graphs, workflows, and agents-as-tools
 - C Requiring every agent to use only one proprietary model provider
 - D Automatically granting every tool unrestricted AWS administrator permissions
 - E Replacing IAM authorization with model reasoning
- 37 When a generative AI model makes something up that may sound plausible and factual but may not be correct, what is it called?
- A Explainability
 - B Fairness
 - C Controllability
 - D Hallucination
- 38 A company wants to implement safeguards for its generative AI application using Amazon Bedrock. Specifically, the company wants to filter undesirable and harmful content as well as redact any personally identifiable information (PII). What do you recommend?
- A Continued pretraining in Amazon Bedrock
 - B Guardrails for Amazon Bedrock
 - C Watermark detection for Amazon Bedrock
 - D Knowledge Bases for Amazon Bedrock
- 39 Which AWS tool provides guidance to help optimize AWS environments for governance, cost savings, performance, security, and fault tolerance in AI systems?
- A AWS Trusted Advisor
 - B AWS Config
 - C AWS Audit Manager
 - D AWS CloudTrail
- 40 Which of the following is correct regarding underfitting/overfitting in machine learning?
- A Underfit models experience high bias, whereas, overfit models experience high variance
 - B Underfit models experience low bias, whereas, overfit models experience high variance
 - C Underfit models experience low bias, whereas, overfit models experience low variance
 - D Underfit models experience high bias, whereas, overfit models experience low variance

- 41 Which of the following are examples of supervised learning? S E L E C T T W O
- A Neural network
 - B Linear regression
 - C Clustering
 - D Document classification
 - E Association rule learning
- 42 Which AWS service provides detailed logs of API calls and user actions for security and operational auditing?
- A AWS CloudTrail
 - B AWS Config
 - C AWS Trusted Advisor
 - D Amazon Inspector
- 43 How would you differentiate between deep learning and traditional machine learning? S E L E C T T W O
- A Deep learning is a subset of machine learning that uses neural networks with many layers to learn from large amounts of data, while traditional machine learning algorithms often require feature extraction and can use various methods such as decision trees or support vector machines
 - B In traditional machine learning, a data scientist manually determines the set of relevant features that the software must analyze, whereas in deep learning, the data scientist gives only raw data to the software and the deep learning network derives the features by itself
 - C Deep learning models are always faster to train than traditional machine learning models, regardless of the dataset size
 - D Traditional machine learning algorithms are only used for supervised learning tasks, whereas deep learning algorithms are only used for unsupervised learning tasks
 - E Deep learning models do not require any data preprocessing, while traditional machine learning models require extensive data preprocessing

- 44 What is the most important security consideration when adding a third-party or automatically loaded tool to a Strands agent?
- A Tools execute with the host process's permissions, so their code and access should be audited and restricted using least privilege
 - B Tool code cannot access files, networks, or operating-system resources
 - C The model automatically converts every unsafe tool into a read-only tool
 - D Registering a tool removes the need for authentication and authorization
- 45 Which of the following are correct regarding model evaluation for Amazon Bedrock? S E L E C T T W O
- A Human model evaluation is valuable for assessing qualitative aspects of the model, whereas, automatic model valuation is valuable for assessing quantitative aspects of the model
 - B Human model evaluation provides model scores that are calculated using various statistical methods such as BERT Score and F1
 - C For human model evaluation, you can use either built-in prompt datasets or your own prompt datasets
 - D Automatic model evaluation is valuable for assessing qualitative aspects of the model, whereas, human model valuation is valuable for assessing quantitative aspects of the model
 - E Automatic model evaluation provides model scores that are calculated using various statistical methods such as BERT Score and F1
- 46 A financial services company applies both supervised and unsupervised learning techniques for fraud detection, wherein, the machine learning solution trains first on the unlabeled data and then on the labeled data. For which of the following categories of Machine Learning would you classify the solution?
- A Transfer Learning
 - B Deep Learning
 - C Self-supervised learning
 - D Semi-supervised learning
- 47 What are the constituents of a good prompting technique?
- A Instructions, Hyperparameters, Input data, Output Indicator
 - B Hyperparameters, Context, Input data, Output Indicator
 - C Instructions, Parameters, Input data, Output Indicator
 - D Instructions, Context, Input data, Output Indicator

- 48 Which prompt engineering technique helps break down a complex problem into smaller logical parts?
- A Negative prompting
 - B Chain-of-thought prompting
 - C Zero shot Prompting
 - D Few shot Prompting
- 49 What is the key difference between SageMaker model cards and AI service cards?
- A SageMaker model cards are used to store data for machine learning models, while AI service cards are used for storing user credentials
 - B SageMaker model cards are used exclusively for monitoring model performance, whereas AI service cards are used for managing model security
 - C SageMaker model cards provide technical documentation for deploying models, while AI service cards offer transparency about the intended use, limitations, and potential impacts of AWS AI services
 - D SageMaker model cards include information about the model such as intended use and risk rating of a model, training details and metrics, evaluation results, and observations. AI service cards provide transparency about AWS AI services' intended use, limitations, and potential impacts
- 50 Which of the following are valid model customization methods for Amazon Bedrock? S E L E C T T W O
- A Fine-tuning
 - B Chain-of-thought prompting
 - C Zero-shot prompting
 - D Continued Pre-training
 - E Retrieval Augmented Generation (RAG)
- 51 Which AWS service is primarily used for governance by providing continuous monitoring and compliance assessment of AWS resource configurations in AI systems?
- A AWS Config
 - B AWS Artifact
 - C Amazon Inspector
 - D AWS Audit Manager

- 52 In the context of secure data engineering for AI systems on AWS, what is the primary difference between data access control and data integrity?
- A Data access control and data integrity are both concerned with encrypting data at rest and in transit
 - B Data access control ensures the accuracy and consistency of data, while data integrity manages who can access the data
 - C Data access control is responsible for data encryption, while data integrity focuses on auditing and logging user activities
 - D Data access control involves authentication and authorization of users, whereas data integrity ensures the data is accurate, consistent, and unaltered
- 53 Which of the following summarizes the capabilities of a multimodal model?
- A A multimodal model can accept a mix of input types such as audio/text, however, it can only create a single type of output
 - B A multimodal model can accept only a single type of input and it can only create a single type of output
 - C A multimodal model can accept only a single type of input, however, it can create a mix of output types such as video/image
 - D A multimodal model can accept a mix of input types such as audio/text and create a mix of output types such as video/image
- 54 A company wants to configure an AI agent to provide real-time currency exchange rates. Which of the following steps is crucial for this setup? S E L E C T T W O
- A Using historical data to predict future rates
 - B Hardcoding exchange rates into the agent
 - C Integrating the agent with a real-time exchange rate API
 - D Ensuring the agent can call relevant Lambda functions
 - E Manually updating the exchange rates daily
- 55 Which of the following aptly summarizes the way diffusion models work?
- A Diffusion models work by training two neural networks in a competitive manner
 - B Diffusion models create new data by iteratively making controlled random changes to an initial data sample
 - C Diffusion models work by learning a compact representation of data called latent space
 - D Diffusion models are a type of transformer-based models that use a self-attention mechanism

- 56 How would you differentiate between overfitting and underfitting in the context of machine learning?
- A Overfitting is desirable as it ensures the model captures all nuances in the training data, while underfitting is desirable as it ensures the model generalizes well to new data
 - B Overfitting and underfitting both refer to a model performing equally well on both the training data and new, unseen data
 - C Overfitting occurs when a model performs well on the training data but poorly on new, unseen data, while underfitting occurs when a model performs poorly on both the training data and new, unseen data
 - D Overfitting occurs when a model is too simple to capture the underlying patterns in the data, while underfitting occurs when a model is too complex and captures noise rather than the actual patterns
- 57 What is the primary function of Foundation Models in Generative AI?
- A They focus exclusively on chatbot interactions.
 - B They are pre-trained on large-scale data to perform multiple tasks, like text generation and summarization.
 - C They are specifically designed for audio synthesis tasks.
 - D They are used only for image generation.
- 58 Which of the following business metrics measures the effectiveness of a generative AI solution in driving user actions like purchases or sign-ups?
- A Average Revenue per User (ARPU)
 - B Efficiency
 - C User satisfaction
 - D Conversion rate
- 59 What does LLM-as-a-judge mean in foundation-model evaluation?
- A A separate evaluator model scores or compares model responses according to defined criteria
 - B The target model trains itself using every answer it generates
 - C A human reviewer manually calculates the model's token usage
 - D A deterministic encryption algorithm checks whether the response is secure

- 60 What is the primary purpose of AWS Audit Manager?
- A Provide on-demand downloads of compliance documents
 - B Continuously scan for software vulnerabilities
 - C Automate evidence collection for compliance audits
 - D Monitor resource configurations
- 61 Which of the following are Machine Learning governance tools that Amazon SageMaker provides?
- A Amazon SageMaker Model Dashboard, Amazon SageMaker Role Manager, Amazon SageMaker Clarify
 - B Amazon SageMaker Role Manager, Amazon SageMaker Model Monitor, Amazon SageMaker Studio
 - C Amazon SageMaker Model Dashboard, Amazon SageMaker Role Manager, Amazon SageMaker Model Monitor
 - D Amazon SageMaker Role Manager, Amazon SageMaker Model Cards, Amazon SageMaker Model Dashboard
- 62 Which of the following correctly distinguishes between tasks performed by Natural Language Processing (NLP) and Computer Vision?
- A NLP is used for tasks such as image recognition and object detection, while Computer Vision is used for text generation and sentiment analysis
 - B NLP and Computer Vision are both used for creating 3D models from textual descriptions
 - C NLP and Computer Vision are both used exclusively for speech recognition tasks
 - D NLP is used for analyzing and generating human language, such as text and speech, while Computer Vision is used for interpreting and understanding visual information from images and videos
- 63 Which of the following is correct regarding the model customization methods for Amazon Bedrock?
- A Continued pre-training uses unlabeled data to pre-train a model, whereas, fine-tuning uses labeled data to train a model
 - B Continued pre-training uses labeled data to pre-train a model, whereas, fine-tuning uses unlabeled data to train a model
 - C Continued pre-training uses labeled data to pre-train a model and fine-tuning also uses labeled data to train a model
 - D Continued pre-training uses unlabeled data to pre-train a model and fine-tuning also uses unlabeled data to train a model

- 64 Which AWS tool is specifically designed to help you continuously audit your AWS usage and automate evidence collection to simplify risk and compliance assessments for AI systems?
- A AWS Audit Manager
 - B AWS Trusted Advisor
 - C AWS Artifact
 - D AWS CloudTrail
- 65 What is the primary role of Model Context Protocol (MCP) in an agentic AI application?
- A It is a model-training algorithm that compresses a teacher model
 - B It is a standardized way for agents to discover and invoke external tools and access resources
 - C It is an AWS pricing model for provisioned inference
 - D It is a benchmark used to compare generated text with a reference answer

RESPONSE SHEET

Mark your answers here
Fill one bubble per question, or two where the question says Select two. Letters
shown in outline only exist for that question – a question with four options has no
E. Score against the key on the next page.

1 D A E	B	C	D	E	2 A B C D E	3 A B C D E	4 A B C D E	5 A B C
6 D A E	B	C	D	E	7 A B C D E	8 A B C D E	9 A B C D E	10 A B C
11 D A E	B	C	D	E	12 A B C D E	13 A B C D E	14 A B C D E	15 A B C
16 D A E	B	C	D	E	17 A B C D E	18 A B C D E	19 A B C D E	20 A B C
21 D A E	B	C	D	E	22 A B C D E	23 A B C D E	24 A B C D E	25 A B C
26 D A E	B	C	D	E	27 A B C D E	28 A B C D E	29 A B C D E	30 A B C
31 D A E	B	C	D	E	32 A B C D E	33 A B C D E	34 A B C D E	35 A B C
36 D A E	B	C	D	E ×2	37 A B C D E	38 A B C D E	39 A B C D E	40 A B C
41 D A E ×2	B	C	D	E ×2	42 A B C D E	43 A B C D E ×2	44 A B C D E	45 A B C
46 D A E ×2	B	C	D	E	47 A B C D E	48 A B C D E	49 A B C D E	50 A B C
51 D A E	B	C	D	E	52 A B C D E	53 A B C D E	54 A B C D E ×2	55 A B C
56 D A E	B	C	D	E	57 A B C D E	58 A B C D E	59 A B C D E	60 A B C
61 D A E	B	C	D	E	62 A B C D E	63 A B C D E	64 A B C D E	65 A B C

FL AGGED FOR REVIEW

Write the number of any question you guessed or want to revisit. These are the ones to
read the explanation for first, even when you happened to get them right.

ANSWER KEY

Score and diagnose
Each row gives the correct letter or letters and the domain the question belongs to.
Tally your domain scores below to see where to study next.

1 A	D2	2 D	D3	3 C	D2	4 B	D2	5 B	D3
6 D	D1	7 C	D3	8 A	D3	9 D	D2	10 A	D3
11 B	D2	12 D	D1	13 A	D2	14 C	D1	15 A	D3
16 A	D3	17 B	D1	18 D	D2	19 C	D4	20 A	D2
21 D	D4	22 D	D3	23 C	D5	24 C	D3	25 B	D2
26 C	D2	27 B	D3	28 C	D3	29 B	D1	30 D	D1
31 D	D5	32 C	D4	33 B	D2	34 D	D1	35 D	D4
36 A, B	D2	37 D	D4	38 B	D3	39 A	D5	40 A	D4
41 A, B	D1	42 A	D5	43 A, B	D1	44 A	D5	45 A, E	D3
46 D	D1	47 D	D3	48 B	D3	49 D	D4	50 A, D	D3
51 A	D5	52 D	D5	53 D	D2	54 C, D	D3	55 B	D2
56 C	D1	57 B	D2	58 D	D2	59 A	D3	60 C	D5
61 D	D4	62 D	D1	63 A	D3	64 A	D5	65 B	D2

DOMAIN		YOUR SCORE	OUT OF	TARGET
D1	Fundamentals of AI and ML		12	9
D2	Fundamentals of Generative AI		17	12
D3	Applications of Foundation Models		19	14
D4	Guidelines for Responsible AI		8	6
D5	Security, Compliance, and Governance		9	6
Total			65	47

AWS scores AIF-C01 on a 100-1000 scale with 700 to pass and does not publish a question-count threshold. The targets above use roughly 72 percent as a working proxy: clear it comfortably in every domain before you sit the real exam.

PA R T T H R E E

Answers explained

Every question, with the correct response, the reasoning behind it, and a primary AWS source you can read afterwards. Work through the ones you missed first, then skim the rest – the distractors are where most of the learning is.

1 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

A developer wants Kiro to access an external issue tracker and internal API through standardized tool interfaces. Which Kiro capability supports this requirement?
A MCP server configuration

Kiro supports MCP servers to connect external tools and data sources to its agent. MCP configuration can be defined at global, project, or custom-agent scope depending on where the integration should be available.

R E F E R E N C E Kiro documentation <https://kiro.dev/docs/>

2 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

A company wants to implement fully managed support for end-to-end Retrieval Augmented Generation (RAG) workflow in Amazon Bedrock. What do you recommend?
D Knowledge Bases for Amazon Bedrock

With Knowledge Bases for Amazon Bedrock, you can give FMs and agents contextual information from your company's private data sources for RAG to deliver more relevant, accurate, and customized responses Knowledge Bases for Amazon Bedrock takes care of the entire ingestion workflow of converting your documents into embeddings (vector) and storing the embeddings in a specialized vector database. Knowledge Bases for Amazon Bedrock supports popular databases for vector storage, including vector engine for Amazon OpenSearch Serverless, Pinecone, Redis Enterprise Cloud, Amazon Aurora (coming soon), and MongoDB (coming soon).

R E F E R E N C E AWS documentation <https://aws.amazon.com/bedrock/knowledge-bases/>

3 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

Which of the following is a generative AI-powered assistant that can answer questions, provide summaries, generate content, and securely complete tasks based on data and information in the enterprise systems?
C Amazon Q Business

Amazon Q Business is a fully managed, generative-AI powered assistant that you can configure to answer questions, provide summaries, generate content, and complete tasks based on your enterprise data. It allows end users to receive immediate, permissions-aware responses from enterprise data sources with citations, for use cases such as IT, HR, and benefits help desks. Amazon Q Business also helps streamline tasks and accelerate problem-solving.

R E F E R E N C E AWS documentation <https://aws.amazon.com/q/>

D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

4

Which of the following represents the correct statement regarding Amazon Q Developer?

- B Amazon Q Developer can be used in integrated development environments (IDEs) as well as the AWS Management Console

You can use Amazon Q Developer in the AWS Management Console, AWS Console Mobile Application, AWS Marketing website, AWS Documentation website, and chat channels integrated with AWS Chatbot to ask questions about AWS. You can ask Amazon Q about AWS architecture, best practices, support, and documentation. Amazon Q can also help with code that you're writing with the AWS SDKs and AWS Command Line Interface (AWS CLI).

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/amazonq/latest/qdeveloper-ug/q-on-aws.html>

D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

5

In foundation-model distillation, what are the roles of the teacher and student models?

- B The teacher is a larger, more capable model whose knowledge is transferred to a smaller, faster student model

Distillation transfers task-relevant behavior from a capable teacher model to a smaller student. The goal is to improve the student's performance for a specific use case while retaining its latency and cost advantages.

R E F E R E N C E Customize a model with distillation in Amazon Bedrock <https://docs.aws.amazon.com/bedrock/latest/>

userguid

e/model-distillation.html

D 1 · F U N D A M E N T A L S O F A I A N D M L

6

What is the key difference between multi-class classification and multi-label classification in machine learning?

- D Multi-class classification assigns each instance to one of several possible classes, while multi-label classification assigns each instance to one or more classes

With multi-label classification, you can train models and classify your documents with more than one label. For example, you can use multi-label classification to categorize customer contact transcripts with one or more labels to identify departments within your company like Payments, Renewals, or Tech Support. These labels can then be mapped to relevant content in your support library or directed towards the appropriate contacts within your company.

R E F E R E N C E AWS documentation [https://aws.amazon.com/blogs/machine-learning/amazon-comprehend-now-supports-m](https://aws.amazon.com/blogs/machine-learning/amazon-comprehend-now-supports-multi-label-custom-classification/)
ulti-label-custom-classification/

7 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

Which of the following highlights the differences between model parameters and hyperparameters in the context of generative AI?

- C Model parameters are values that define a model and its behavior in interpreting input and generating responses. Hyperparameters are values that can be adjusted for model customization to control the training process

Hyperparameters are values that can be adjusted for model customization to control the training process and, consequently, the output custom model. In other words, hyperparameters are external configurations set before the training process begins. They control the training process and the structure of the model but are not adjusted by the training algorithm itself.

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/bedrock/latest/userguide/key-definitions.html>

8 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

A company is using Amazon Bedrock and it wants to set an upper limit on the number of tokens returned in the model's response. Which of the following inference parameters would you recommend for the given use case?

- A Response length

The response-length or maximum-output-token setting places an upper bound on generated tokens; it can truncate output and is not the same as asking for concise writing.

R E F E R E N C E Amazon Bedrock inference parameters <https://docs.aws.amazon.com/bedrock/latest/userguide/inference-parameters.html>

9 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

Which feature of Amazon Bedrock allows developers to automate repetitive tasks in their AI applications?

- D Use of agents

Agents enable an application to orchestrate steps and invoke actions or tools, which is the relevant Bedrock capability for automating tasks.

R E F E R E N C E What is Amazon Bedrock? <https://docs.aws.amazon.com/bedrock/latest/userguide/what-is-bedrock.html>

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D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

An ecommerce company is transitioning to AWS Cloud and wants to use Amazon Bedrock for product recommendations. The company wants to provide its own labeled training dataset to improve the selected Foundation Model's (FM) performance. Which of the following represents the best-fit solution for the given use case?

- A Leverage Amazon Bedrock to make a separate copy of the base FM model and train this private copy of the model using the labeled training dataset

Bedrock customization creates a customer-specific copy of a supported base model and trains that copy with the customer dataset; the shared base model itself is not retrained.

R E F E R E N C E What is Amazon Bedrock? <https://docs.aws.amazon.com/bedrock/latest/userguide/what-is-bedrock.html>

11

D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

Which of the following can assist IT professionals in coding, testing, and upgrading applications?

- B Amazon Q Developer

Amazon Q Developer is a generative artificial intelligence (AI) powered conversational assistant that can help you understand, build, extend, and operate AWS applications. You can ask questions about AWS architecture, your AWS resources, best practices, documentation, support, and more. Amazon Q is constantly updating its capabilities so your questions get the most contextually relevant and actionable answers.

R E F E R E N C E AWS documentation <https://aws.amazon.com/q/>

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D 1 · F U N D A M E N T A L S O F A I A N D M L

What is a key difference between structured data and unstructured data in the context of Machine Learning?

- D Structured data is organized in a predefined manner, often in rows and columns, making it easy to search and analyze, while unstructured data lacks a specific format and includes data like text, images, and videos

Structured data is highly organized and formatted, typically found in databases and spreadsheets, making it straightforward to search and analyze. Unstructured data, on the other hand, does not have a predefined format and can include diverse data types such as text, images, and videos.

R E F E R E N C E AWS documentation <https://aws.amazon.com/compare/the-difference-between-structured-data-and-unstructured-data/>

- 13
- D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- Which pricing model is suitable for large and steady workloads in Amazon Bedrock?
A Provisioned throughput mode
- Provisioned Throughput reserves model capacity and is intended for large, predictable, steady workloads that need consistent throughput.
R E F E R E N C E What is Amazon Bedrock? <https://docs.aws.amazon.com/bedrock/latest/userguide/what-is-bedrock.html>
- 14
- D 1 · F U N D A M E N T A L S O F A I A N D M L
- What is one of the primary challenges in machine learning implementation?
C Difficulty in collecting and preparing high-quality data for training models
- One of the main challenges in machine learning implementation is the difficulty in collecting and preparing high-quality data for training models. High-quality data is essential for building effective machine learning models, and ensuring that the data is clean, relevant, and well-prepared can be a complex and time-consuming process.
R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/machine-learning/>
- 15
- D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- What is the primary function of a prompt in the context of large language models?
A To serve as the input or query that generates a response from the model
- A prompt is the input that gives a model instructions, context, and data from which it generates a response.
R E F E R E N C E Amazon Bedrock Prompt management <https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-management.html>
- 16
- D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- Which feature of Amazon Q Business allows the AI assistant to interact with third-party applications like Jira and Salesforce?
A Plugins
- Amazon Q Business plugins allow the assistant to perform supported actions in third-party systems such as Jira and Salesforce.
R E F E R E N C E What is Amazon Q Business? <https://docs.aws.amazon.com/amazonq/latest/qbusiness-ug/what-is.html>

- 17 D 1 · F U N D A M E N T A L S O F A I A N D M L
- A company is creating a custom search solution that will bring together the company's data repositories, FAQs, and support tickets. The support tickets might contain personally identifiable information (PII) that needs to be redacted before the tickets are processed to create the search indexes. Which AWS service will help you redact the PII in support tickets?
- B Amazon Comprehend
- Amazon Comprehend is a natural language processing (NLP) service that uses machine learning to discover insights from text. Amazon Comprehend provides Custom Entity Recognition, Custom Classification, Key Phrase Extraction, Sentiment Analysis, Entity Recognition, and more APIs so you can easily integrate natural language processing into your applications. You simply call the Amazon Comprehend APIs in your application and provide the location of the source document or text.
R E F E R E N C E AWS documentation <https://aws.amazon.com/comprehend/>
- 18 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- What is the primary benefit of using Amazon Bedrock for AI and machine learning tasks?
- D It allows users to access and use pre-trained foundation models from various AI model providers without needing to manage infrastructure.
- Amazon Bedrock provides managed access to foundation models from multiple providers without requiring customers to operate the underlying model infrastructure.
R E F E R E N C E What is Amazon Bedrock? <https://docs.aws.amazon.com/bedrock/latest/userguide/what-is-bedrock.html>
- 19 D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I
- What is the primary purpose of AWS AI service cards?
- C To offer transparency and information about the intended use, limitations, and potential impacts of AWS AI services, helping users implement Responsible AI practices
- AWS AI service cards are designed to offer transparency and provide users with crucial information about the intended use, limitations, and potential impacts of AWS AI services. This helps users make informed decisions and implement Responsible AI practices by understanding the implications of using these services.
R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/machine-learning/introducing-aws-ai-service-cards-a-new-resource-to-enhance-transparency-and-advance-responsible-ai/>

- 20 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- What is a key role of agents in generative AI applications?
- A To act as intermediaries between the AI model and business backend systems
- Agents connect a foundation model to tools, APIs, data, and business systems so the application can plan and take actions instead of only producing text.
- R E F E R E N C E AIF-C01 Domain 2 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain2.html>
- 21 D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I
- Which of the following scenarios best illustrates human bias in the context of machine learning?
- D A data scientist selects features for a machine learning model based on their personal beliefs about which attributes are important, leading to a biased model
- This scenario exemplifies human bias, where the data scientist's personal beliefs influence the feature selection process, potentially leading to a biased machine learning model.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/machine-learning/responsible-ai/>
- 22 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- What is a key difference between Foundation Models (FMs) and Large Language Models (LLMs) in the context of generative AI?
- D Foundation Models serve as a broad base for various AI applications by providing generalized capabilities, whereas Large Language Models are specialized for understanding and generating human language
- Foundation Models provide a broad base with generalized capabilities that can be applied to various tasks such as natural language processing (NLP), question answering, and image classification. The size and general-purpose nature of FMs make them different from traditional ML models, which typically perform specific tasks, like analyzing text for sentiment, classifying images, and forecasting trends. Generally, an FM uses learned patterns and relationships to predict the next item in a sequence.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/generative-ai/>

23 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E

A bank wants refund limits to be enforced even if an agent is manipulated by a malicious prompt. Which design provides the strongest deterministic control?

C Attach an AgentCore policy engine with a Cedar policy to the Gateway

Policy in AgentCore evaluates tool requests at the Gateway boundary, outside the agent's reasoning code. Cedar policies provide deterministic authorization and can restrict tool access using identity and tool-input conditions.

R E F E R E N C E Policy in Amazon Bedrock AgentCore <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/cy.html>

24 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

How does the inference parameter Top K influence the model response for Amazon Bedrock?

C Influences the number of most-likely candidates that the model considers for the next token

The inference parameter Top K represents the number of most likely candidates that the model considers for the next token. Choose a lower value to decrease the size of the pool and limit the options to more likely outputs. Choose a higher value to increase the size of the pool and allow the model to consider less likely outputs.

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/bedrock/latest/userguide/inference-parameters.html>

25 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

Which statement best describes Amazon Bedrock AgentCore?

B It is a modular platform for securely building, deploying, and operating agents at scale with different frameworks and models

AgentCore provides modular capabilities for running, connecting, securing, remembering, observing, and evaluating agents. Its services can be used together or independently and support multiple agent frameworks and foundation models.

R E F E R E N C E What is Amazon Bedrock AgentCore? <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/dhistory.html>

26 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

What is the primary purpose of vectors in large language models?

C To represent the numerical meaning of words

Vectors are numerical representations that encode semantic properties, allowing models and retrieval systems to compare meaning mathematically.

R E F E R E N C E AIF-C01 Domain 2 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain2.html>

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D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

In Amazon Bedrock AgentCore Gateway, what does a gateway target represent?

B The backend tool or resource and the configuration Gateway uses to connect to it

A target defines the backend connected to a gateway. Depending on its category, a target can represent MCP capabilities, an HTTP service, or an inference endpoint, together with the connection and authorization configuration.

R E F E R E N C E AgentCore Gateway core concepts [https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gat](https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway-core-concepts.html)

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D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

The marketing department at a retail company is working on a campaign for the upcoming Thanksgiving holidays. The department wants to exclude competitive brand names or sensitive topics from the content produced via generative AI prompts. What type of prompting technique does the given use case represent?

C Negative prompting

Negative prompting states content or behavior that the model should avoid, such as competitor names or sensitive topics.

R E F E R E N C E Amazon Bedrock Prompt management [https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-mana](https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-management.html)

gement.html

29

D 1 · F U N D A M E N T A L S O F A I A N D M L

What is the key difference between machine learning and artificial intelligence?

B Machine learning is a subset of artificial intelligence that involves training algorithms to learn from data, while artificial intelligence encompasses a wider range of technologies aimed at simulating human intelligence

Artificial intelligence is an umbrella term for different strategies and techniques used to make machines more human-like. AI includes everything from smart assistants like Alexa, chatbots, and image generators to robotic vacuum cleaners and self-driving cars. In contrast, machine learning is a subset of artificial intelligence, focusing specifically on training algorithms to learn from data and make predictions or decisions.

R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/machine-learning/>

- 30 D 1 · F U N D A M E N T A L S O F A I A N D M L
- How can you prevent model-overfitting in machine learning?
- D By using techniques such as cross-validation, regularization, and pruning to simplify the model and improve its generalization
- To prevent overfitting, techniques such as cross-validation, regularization, and pruning are employed. Cross-validation helps ensure the model generalizes well to unseen data by dividing the data into multiple training and validation sets. Regularization techniques, such as L1 and L2 regularization, penalize complex models to reduce overfitting.
- R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/machine-learning/latest/dg/model-fit-underfitting-vs-overfitting.html>
- 31 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E
- Which task ensures the confidentiality and integrity of data at rest and in transit in AI systems?
- D Data encryption
- Encryption protects data confidentiality and integrity at rest and in transit; it complements, rather than replaces, access control and monitoring.
- R E F E R E N C E AIF-C01 Domain 5 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-01/ai-practitioner-01-domain5.html>
- 32 D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I
- Which AWS service offers flexibility to include human input across the Machine Learning lifecycle to improve the accuracy and relevancy of the models?
- C Amazon SageMaker Ground Truth
- Human-in-the-loop is the process of harnessing human input across the ML lifecycle to improve the accuracy and relevancy of models. Humans can perform a variety of tasks, from data generation and annotation, to model review and customization. Human intervention is especially important for generative AI applications, where humans are typically both the requester and consumer of the content.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/sagemaker/groundtruth/>

- 33 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- A company wants its agents to use existing AWS Lambda functions and REST APIs without rewriting each integration for a different agent framework. Which solution best meets this requirement?
- B Add the functions and APIs as targets behind an Amazon Bedrock AgentCore Gateway
- AgentCore Gateway can expose Lambda functions and REST APIs as MCP-compatible tools through a unified endpoint. Gateway handles the integration schema, authentication configuration, and routing to the configured targets.
- R E F E R E N C E Set up an AgentCore gateway <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway-building.html>
- b
- 34 D 1 · F U N D A M E N T A L S O F A I A N D M L
- A business needs an automated solution that can extract text from thousands of receipts and invoices generated across all its stores. Which AWS Machine Learning (ML) service can offer the most optimal solution for this use case ?
- D Amazon Textract
- Amazon Textract is a machine learning (ML) service that automatically extracts text, handwriting, layout elements, and data from scanned documents. It goes beyond simple optical character recognition (OCR) to identify, understand, and extract specific data from documents. Today, many companies manually extract data from scanned documents such as PDFs, images, tables, and forms, or through simple OCR software that requires manual configuration (which often must be updated when the form changes).
- R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/textract/latest/dg/what-is.html>
- 35 D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I
- Which Amazon SageMaker service will help understand how an input feature contributes to the predictions of a machine learning model?
- D Amazon SageMaker Clarify
- SageMaker Clarify automatically evaluates foundation models for your generative AI use case with metrics such as accuracy, robustness, and toxicity to support your responsible AI initiative. SageMaker Clarify explains how input features contribute to your model predictions during model development and inference. Evaluate your FM during customization using automatic and human-based evaluations.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/sagemaker/clarify/>

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D 2 · FUNDAMENTALS OF GENERATIVE AI

Which capabilities are supported by the Strands Agents SDK?

- A + B Connecting to tools through MCP; Building multi-agent patterns such as graphs, workflows, and agents-as-tools

Strands supports MCP tools and several multi-agent patterns. It is model-provider flexible, but developers remain responsible for configuring permissions and safely governing the tools an agent can execute.

REFERENCE Get started with Strands Agents <https://strandsagents.com/docs/user-guide/quickstart/overview/>

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D 4 · GUIDELINES FOR RESPONSIBLE AI

When a generative AI model makes something up that may sound plausible and factual but may not be correct, what is it called?

- D Hallucination

Generative AI models are as reliable as the data they're trained on and can access. There is a risk of hallucination, which is when the models make something up that may sound plausible and factual but which may not be correct. Any organizations or teams that use generative AI to make decisions or prioritize actions, must build responsible AI systems that are fair, explainable, robust, secure, transparent, and that safeguard privacy.

REFERENCE AWS documentation <https://aws.amazon.com/blogs/publicsector/generative-ai-understand-the-challenges-to-realize-the-opportunities/>

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D 3 · APPLICATIONS OF FOUNDATION MODELS

A company wants to implement safeguards for its generative AI application using Amazon Bedrock. Specifically, the company wants to filter undesirable and harmful content as well as redact any personally identifiable information (PII). What do you recommend?

- B Guardrails for Amazon Bedrock

Guardrails for Amazon Bedrock help you implement safeguards for your generative AI applications based on your use cases and responsible AI policies. Guardrails for Amazon Bedrock helps control the interaction between users and FMs by filtering undesirable and harmful content and will soon redact personally identifiable information (PII), enhancing content safety and privacy in generative AI applications. You cannot use guardrails to implement RAG workflow in Amazon Bedrock.

REFERENCE AWS documentation <https://aws.amazon.com/bedrock/guardrails/>

- 39 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E
- Which AWS tool provides guidance to help optimize AWS environments for governance, cost savings, performance, security, and fault tolerance in AI systems?
A AWS Trusted Advisor
- AWS Trusted Advisor provides recommendations across cost optimization, performance, security, fault tolerance, and service limits.
R E F E R E N C E AWS Trusted Advisor <https://docs.aws.amazon.com/awssupport/latest/user/trusted-advisor.html>
- 40 D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I
- Which of the following is correct regarding underfitting/overfitting in machine learning?
A Underfit models experience high bias, whereas, overfit models experience high variance
- Underfitting is associated with high bias; overfitting is associated with high variance and poor generalization to unseen data.
R E F E R E N C E AIF-C01 Domain 4 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain4.html>
- 41 D 1 · F U N D A M E N T A L S O F A I A N D M L
- Which of the following are examples of supervised learning?
A + B Neural network; Linear regression
- Supervised learning algorithms train on sample data that specifies both the algorithm's input and output. For example, the data could be images of handwritten numbers that are annotated to indicate which numbers they represent. Given sufficient labeled data, the supervised learning system would eventually recognize the clusters of pixels and shapes associated with each handwritten number.
R E F E R E N C E AWS documentation <https://aws.amazon.com/compare/the-difference-between-machine-learning-supervised-and-unsupervised/>
- 42 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E
- Which AWS service provides detailed logs of API calls and user actions for security and operational auditing?
A AWS CloudTrail
- AWS CloudTrail records AWS API activity and identity context for auditing, investigations, and operational analysis.
R E F E R E N C E What is AWS CloudTrail? <https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudtrail-use-guide.html>

D 1 · F U N D A M E N T A L S O F A I A N D M L

43

How would you differentiate between deep learning and traditional machine learning?

- A + B Deep learning is a subset of machine learning that uses neural networks with many layers to learn from large amounts of data, while traditional machine learning algorithms often require feature extraction and can use various methods such as decision trees or support vector machines; In traditional machine learning, a data scientist manually determines the set of relevant features that the software must analyze, whereas in deep learning, the data scientist gives only raw data to the software and the deep learning network derives the features by itself

Deep learning is a subset of machine learning that employs neural networks with multiple layers (hence the term "deep") to automatically learn representations from large datasets. Traditional machine learning, on the other hand, often involves manual feature extraction and can use a variety of algorithms like decision trees, support vector machines, and linear regression. Traditional machine learning methods require human input for the machine learning software to work sufficiently well.

R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/artificial-intelligence/>

D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E

44

What is the most important security consideration when adding a third-party or automatically loaded tool to a Strands agent?

- A Tools execute with the host process's permissions, so their code and access should be audited and restricted using least privilege

Agent tools can create real-world side effects and execute with the permissions available to their host process. Teams should review tool behavior, trust only appropriate code, and apply least-privilege file, network, and AWS permissions.

R E F E R E N C E Strands tools overview <https://strandsagents.com/docs/user-guide/concepts/tools/>

D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

45

Which of the following are correct regarding model evaluation for Amazon Bedrock?

- A + E Human model evaluation is valuable for assessing qualitative aspects of the model, whereas, automatic model valuation is valuable for assessing quantitative aspects of the model; Automatic model evaluation provides model scores that are calculated using various statistical methods such as BERT Score and F1

Model evaluation is the process of evaluating and comparing model outputs to determine the model that is best suited for a use case. You can choose to create either an automatic model evaluation job or a model evaluation job that uses a human workforce. Automatic model evaluation jobs allow you to quickly evaluate a model's ability to perform a task.

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/bedrock/latest/userguide/model-evaluation.html>

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D 1 · F U N D A M E N T A L S O F A I A N D M L

A financial services company applies both supervised and unsupervised learning techniques for fraud detection, wherein, the machine learning solution trains first on the unlabeled data and then on the labeled data. For which of the following categories of Machine Learning would you classify the solution?

D Semi-supervised learning

Semi-supervised learning is when you apply both supervised and unsupervised learning techniques to a common problem. This technique relies on using a small amount of labeled data and a large amount of unlabeled data to train systems. First, the labeled data is used to partially train the machine learning algorithm.

R E F E R E N C E AWS documentation <https://aws.amazon.com/compare/the-difference-between-machine-learning-supervised-and-unsupervised/>

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D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

What are the constituents of a good prompting technique?

D Instructions, Context, Input data, Output Indicator

The following are the constituents of a good prompting technique: • Instructions – a task for the model to do (description, how the model should perform) • Context – external information to guide the model • Input data – the input for which you want a response • Output Indicator – the output type or format

R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/prompt-engineering/>

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D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

Which prompt engineering technique helps break down a complex problem into smaller logical parts?

B Chain-of-thought prompting

Chain-of-thought prompting is a technique that breaks down a complex question into smaller, logical parts that mimic a train of thought. This helps the model solve problems in a series of intermediate steps rather than directly answering the question. This enhances its reasoning ability.

R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/prompt-engineering/>

D 4 · GUIDELINES FOR RESPONSIBLE AI

49

What is the key difference between SageMaker model cards and AI service cards?

- D SageMaker model cards include information about the model such as intended use and risk rating of a model, training details and metrics, evaluation results, and observations. AI service cards provide transparency about AWS AI services' intended use, limitations, and potential impacts

You can use Amazon SageMaker Model Cards to document critical details about your machine learning (ML) models in a single place for streamlined governance and reporting. You can catalog details such as the intended use and risk rating of a model, training details and metrics, evaluation results and observations, and additional call-outs such as considerations, recommendations, and custom information.

R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/machine-learning/introducing-aws-ai-service-cards-a-new-resource-to-enhance-transparency-and-advance-responsible-ai/>

D 3 · APPLICATIONS OF FOUNDATION MODELS

50

Which of the following are valid model customization methods for Amazon Bedrock?

- A + D Fine-tuning; Continued Pre-training

Model customization involves further training and changing the weights of the model to enhance its performance. You can use continued pre-training or fine-tuning for model customization in Amazon Bedrock. In the continued pre-training process, you provide unlabeled data to pre-train a foundation model by familiarizing it with certain types of inputs.

R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/machine-learning/best-practices-to-build-generative-ai-applications-on-aws/>

D 5 · SECURITY, COMPLIANCE, AND GOVERNANCE

51

Which AWS service is primarily used for governance by providing continuous monitoring and compliance assessment of AWS resource configurations in AI systems?

- A AWS Config

AWS Config records resource configurations and evaluates them against Config rules for continuous compliance assessment.

R E F E R E N C E What is AWS Config? <https://docs.aws.amazon.com/config/latest/developerguide/WhatIsConfig.html>

52 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E

In the context of secure data engineering for AI systems on AWS, what is the primary difference between data access control and data integrity?

D Data access control involves authentication and authorization of users, whereas data integrity ensures the data is accurate, consistent, and unaltered

Data access control is about managing who can access data and what actions they can perform, typically through mechanisms like authentication and authorization. Data integrity, on the other hand, focuses on maintaining the accuracy, consistency, and trustworthiness of data throughout its lifecycle, ensuring that data remains unaltered and accurate during storage, processing, and transmission.

R E F E R E N C E AWS documentation <https://aws.amazon.com/ai/generative-ai/security/>

53 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

Which of the following summarizes the capabilities of a multimodal model?

D A multimodal model can accept a mix of input types such as audio/text and create a mix of output types such as video/image

A multimodal model is an artificial intelligence system designed to process and understand multiple types of data, such as text, images, audio, and video. Unlike unimodal models, which handle a single type of data, multimodal models can integrate and make sense of information from various sources, allowing them to perform more complex and versatile tasks. Multimodal models represent a significant advancement in AI, enabling the integration and understanding of multiple types of data.

R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/industries/multimodal-data-analysis-with-aws-health-and-machine-learning-services/>

54 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

A company wants to configure an AI agent to provide real-time currency exchange rates. Which of the following steps is crucial for this setup?

C + D Integrating the agent with a real-time exchange rate API; Ensuring the agent can call relevant Lambda functions

A real-time data source requires an API integration, and an agent action can invoke a Lambda function that calls that API.

R E F E R E N C E How Amazon Bedrock Agents works [https://docs.aws.amazon.com/bedrock/latest/userguide/agents-how.h](https://docs.aws.amazon.com/bedrock/latest/userguide/agents-how.html)

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- 55 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- Which of the following aptly summarizes the way diffusion models work?
- B Diffusion models create new data by iteratively making controlled random changes to an initial data sample
- Diffusion models work by first corrupting data with noise through a forward diffusion process and then learning to reverse this process to denoise the data. They use neural networks to predict and remove the noise step by step, ultimately generating new, structured data from random noise.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/generative-ai/>
- 56 D 1 · F U N D A M E N T A L S O F A I A N D M L
- How would you differentiate between overfitting and underfitting in the context of machine learning?
- C Overfitting occurs when a model performs well on the training data but poorly on new, unseen data, while underfitting occurs when a model performs poorly on both the training data and new, unseen data
- Overfitting happens when a model learns the training data too well, including noise and outliers, leading to excellent performance on the training data but poor generalization to new, unseen data. Underfitting occurs when a model is too simplistic to capture the underlying patterns in the data, resulting in poor performance on both the training data and new data.
- R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/machine-learning/latest/dg/model-fit-underfitting-vs-overfitting.html>
- 57 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- What is the primary function of Foundation Models in Generative AI?
- B They are pre-trained on large-scale data to perform multiple tasks, like text generation and summarization.
- Foundation models are pretrained at scale and can support multiple downstream tasks, including generation and summarization, rather than only one modality or use case.
- R E F E R E N C E What is Amazon Bedrock? <https://docs.aws.amazon.com/bedrock/latest/userguide/what-is-bedrock.html>
- 58 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- Which of the following business metrics measures the effectiveness of a generative AI solution in driving user actions like purchases or sign-ups?
- D Conversion rate
- Conversion rate measures the percentage of users who complete a desired action, such as a purchase or sign-up, and therefore directly measures action-driving effectiveness.
- R E F E R E N C E AIF-C01 Domain 2 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain2.html>
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- 59 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- What does LLM-as-a-judge mean in foundation-model evaluation?
- A A separate evaluator model scores or compares model responses according to defined criteria
- LLM-as-a-judge uses an evaluator foundation model to assess another model's output. The evaluation can use built-in or custom instructions and metrics, and can consider a reference response when one is provided.
- R E F E R E N C E Amazon Bedrock judge-based evaluation metrics <https://docs.aws.amazon.com/bedrock/latest/userguide/el-evaluation-metrics.html>
- 60 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E
- What is the primary purpose of AWS Audit Manager?
- C Automate evidence collection for compliance audits
- AWS Audit Manager automates evidence collection for audits. AWS Artifact supplies compliance reports, Inspector finds vulnerabilities, and Config assesses resource configuration.
- R E F E R E N C E What is AWS Audit Manager? <https://docs.aws.amazon.com/audit-manager/latest/userguide/what-is.htm>
- 61 D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I
- Which of the following are Machine Learning governance tools that Amazon SageMaker provides?
- D Amazon SageMaker Role Manager, Amazon SageMaker Model Cards, Amazon SageMaker Model Dashboard
- SageMaker provides purpose-built ML governance tools across the ML lifecycle. With Amazon SageMaker Role Manager, administrators can define minimum permissions in minutes. Amazon SageMaker Model Cards make it easier to capture, retrieve, and share essential model information from conception to deployment, and Amazon SageMaker Model Dashboard keeps you informed on production model behavior, all in one place.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/sagemaker/ml-governance/>
- 62 D 1 · F U N D A M E N T A L S O F A I A N D M L
- Which of the following correctly distinguishes between tasks performed by Natural Language Processing (NLP) and Computer Vision?
- D NLP is used for analyzing and generating human language, such as text and speech, while Computer Vision is used for interpreting and understanding visual information from images and videos
- Natural Language Processing (NLP) focuses on tasks involving human language, such as text analysis, speech recognition, language translation, and sentiment analysis. Computer Vision, on the other hand, deals with visual information, enabling tasks like image recognition, object detection, image segmentation, and video analysis.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/artificial-intelligence/>

D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

63

Which of the following is correct regarding the model customization methods for Amazon Bedrock?

- A Continued pre-training uses unlabeled data to pre-train a model, whereas, fine-tuning uses labeled data to train a model

Continued pre-training generally uses unlabeled domain data, while supervised fine-tuning uses labeled prompt-response or task examples.

R E F E R E N C E What is Amazon Bedrock? <https://docs.aws.amazon.com/bedrock/latest/userguide/what-is-bedrock.html>

D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E

64

Which AWS tool is specifically designed to help you continuously audit your AWS usage and automate evidence collection to simplify risk and compliance assessments for AI systems?

- A AWS Audit Manager

AWS Audit Manager automates evidence collection and helps map AWS usage to audit controls and compliance frameworks.

R E F E R E N C E What is AWS Audit Manager? <https://docs.aws.amazon.com/audit-manager/latest/userguide/what-is.htm>

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D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

65

What is the primary role of Model Context Protocol (MCP) in an agentic AI application?

- B It is a standardized way for agents to discover and invoke external tools and access resources

MCP standardizes how AI applications connect to capabilities such as tools, prompts, and resources. This reduces the need to build a different proprietary integration for every agent and external system.

R E F E R E N C E Use an AgentCore gateway <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway-us>

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